



Computer Networks Week # 15

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Application Layer

- Provides applications with the ability to access the services of the other layers
- New protocols and services are always being developed in this category
- Some core application layer protocols
 - HTTP
 - FTP
 - Telnet
 - SMTP
 - POP3
 - IMAP
 - SNMP etc.

Hyper Text Transfer Protocol (HTTP)

- Protocol relating to web applications
 - Foundation of data communication for world wide web
- Functions as a request-response protocol in client-server computing model
 - Web browser may be a client and
 - An application running on a computer which hosts website may be a server
- Upload information to the server
 - Etc.
- Default port number is 80

File Transfer Protocol (FTP)

- Designed for faster file transfer over the Internet compared to using the HTTP protocol
 - Used for downloading from most MP3 sites,

For example

- FTP sites can be configured alongside a web site to support FTP file transfer
- FTP default ports are 20 and 21

HTTP and FTP

- File transfer under FTP is faster than file transfer under HTTP
- Choose an FTP site if there is one for downloading files etc.

Simple Mail Transfer Protocol (SMTP)

- Governs the transmission of mail messages and attachments
- SMTP is used in the case of outgoing messages
- More powerful protocols such as POP3 and IMAP4 are needed and available to manage incoming messages

POP3/IMAP4

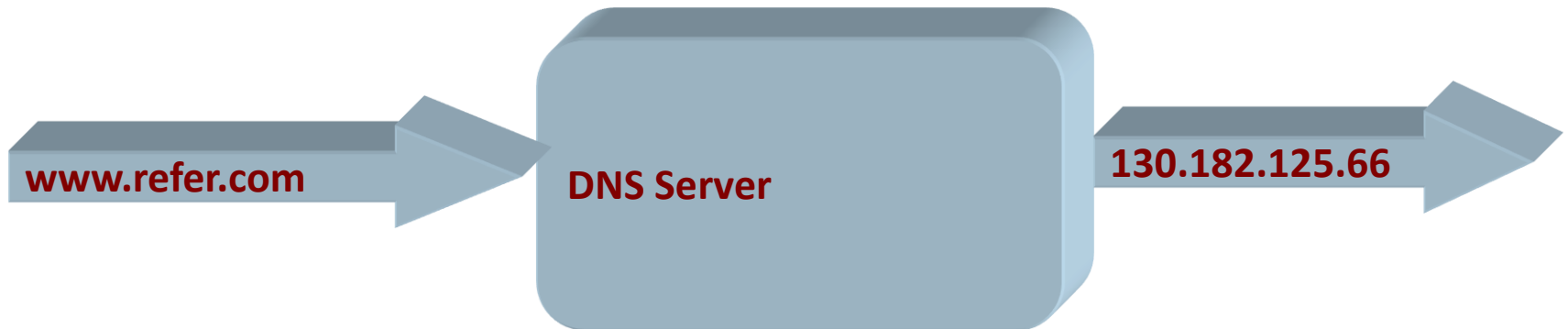
- Used for incoming mail from remote server to a local email client
- Post Office Protocol version 3 (POP3) is the older protocol
- Internet Message Access Protocol (IMAP4) is the more advanced protocol

Telnet

- Supports terminal emulation or host sessions
- For example, Telnet can be used for accessing a Unix machine and emulating a terminal attached to the Unix computer

Domain Name System (DNS)

Resolves domain names to IP addresses
and vice versa



Simple Network Management Protocol (SNMP)

Facilitates the management of SNMP compliant routers, bridges, switches etc. by enabling the collection and exchange of network management information.

